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Scientific Computing

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2.5: Grönwall lemmas and their use for Euler

Outline

Lecture 2.5: Grönwall lemmas and their use for Euler

- Grönwall lemma, continuous case
- Grönwall lemma, discrete case
- Application to the explicit Euler error recurrence
- Q & A

Series: 2.1 Euler schemes. 2.2 errors. 2.3 geometry. 2.4 explicit convergence. 2.5 Grönwall. 2.6 implicit convergence.

Lemma: continuous Grönwall inequality

Lemma 2.1 (Continuous Grönwall lemma: $u \le A + \int B\,u \;\Rightarrow\; u \le Ae^{B(t-t_0)}$)

Let u be continuous on $[t_0, T]$ and let $A, B \ge 0$ be constants with

$$u(t) \le A + \int_{t_0}^{t} B\,u(s)\,ds.$$

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Then

$$u(t) \le A\,e^{B(t-t_0)}.$$

Proof of the continuous Grönwall lemma, part 1

Proof of continuous Grönwall lemma, part 1: differential inequality for $w = ve^{-Bt}$.

1. Let $v(t) = \int_{t_0}^{t} B\,u(s)\,ds$. Then $v' = Bu$, $v(t_0) = 0$ and $u \leq A + v$.

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2. Let $w(t) = v(t)e^{-Bt}$. Its derivative is

$$w'(t) = Bu(t)e^{-Bt} - Bv(t)e^{-Bt} = Be^{-Bt}\big(u(t) - v(t)\big).$$

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3. Since $u - v \leq A$ and $B \geq 0$:

$$w'(t) \leq BAe^{-Bt}.$$

□

Proof of the continuous Grönwall lemma, part 2

Proof of continuous Grönwall lemma, part 2: integrating the inequality.

1. Integrate from t_0 to t and use $w(t_0) = 0$:

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3. Substitute into $u \leq A + v$:

$$u(t) \leq A + A\big(e^{B(t-t_0)} - 1\big) = A\, e^{B(t-t_0)}.$$



Lemma: discrete Grönwall inequality

Lemma 2.2 (Discrete Grönwall lemma: $u_n \le A + B\sum_{k<n} u_k \;\Rightarrow\; u_n \le A(1+B)^n$)

Let $A, B \ge 0$ and let $(u_n)_{n\ge 0}$ satisfy

$$u_n \le A + B\sum_{k=0}^{n-1} u_k \qquad \text{for all } n \ge 0.$$

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Then

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Proof of the discrete Grönwall lemma, part 1

Proof of discrete Grönwall lemma, part 1: reduction to partial sums.

- Let $v_n = \sum_{k=0}^{n-1} u_k$, so $v_0 = 0$ and $u_n \leq A + Bv_n$.

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- Let $v_n = \sum_{k=0}^{n-1} u_k$, so $v_0 = 0$ and $u_n \leq A + Bv_n$.
- Sum the inequality $u_k \leq A + Bv_k$ over $k = 0, \ldots, n-1$:

$$v_n \leq nA + B \sum_{k=0}^{n-1} v_k.$$

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- Claim, to be proved by induction:

$$v_n \leq \frac{A}{B}\Big((1+B)^n - 1\Big).$$



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Proof of discrete Grönwall lemma, part 2: induction on partial sums.

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$$\sum_{k=0}^{n-1} v_k \leq \frac{A}{B} \sum_{k=0}^{n-1} \left((1+B)^k - 1\right) = \frac{A}{B}\left(\frac{(1+B)^n - 1}{B} - n\right).$$

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- Insert into $v_n \leq nA + B\sum v_k$:

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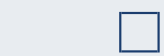
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- Finally $u_n \leq A + Bv_n \leq A + A\big((1+B)^n - 1\big) = A(1+B)^n$.



Remark: general forms of the Grönwall lemma

Remark 2.3 (Grönwall lemmas with variable coefficients $a(t), b(t)$ and α_k)

Continuous. If $b \geq 0$ and $u(t) \leq a(t) + \int_0^t b(s)u(s)\,ds$, then with $\mathcal{B}(t) = \int_0^t b$:

$$u(t) \leq a(t) + \int_0^t a(s)b(s)e^{\mathcal{B}(t)-\mathcal{B}(s)}ds.$$

Discrete. If $A_n = \sum_{k<n} \alpha_k$ with $\alpha_k \geq 0$ and $e_n \leq A_n + B\sum_{k<n} e_k$, then

$$e_n \leq A_n(1+B)^n.$$

Problem: apply discrete Grönwall to explicit Euler

Problem 2.4 (Bounding the explicit Euler error recurrence with the discrete Grönwall lemma)

From the convergence proof:

$$\|e_{n+1}\| \leq (1 + hL)\|e_n\| + \|\tau_{n+1}\|, \qquad e_0 = 0.$$

Bound $\|e_n\|$ with the discrete Grönwall lemma.

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Step 1: bring the recurrence to Grönwall form

Subtract $\|e_n\|$ and sum over $k = 0, \dots, n-1$ (telescoping, $e_0 = 0$):

$$\|e_n\| \le \sum_{k=0}^{n-1} \|\tau_{k+1}\| + hL \sum_{k=0}^{n-1} \|e_k\|.$$

The Grönwall constant is $B = hL$ (not $1 + hL$).

Solution: bound from the discrete Grönwall lemma

Step 2: bound the sum of local errors

$\|\tau_{k+1}\| \leq Mh^2/2$ and $nh \leq \beta - \alpha$, so

$$\sum_{k=0}^{n-1} \|\tau_{k+1}\| \leq \frac{nMh^2}{2} \leq \frac{(\beta-\alpha)M}{2}\, h =: A.$$

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Step 3: apply the lemma with A and $B = hL$

$$\|e_n\| \le A(1+hL)^n \le A\, e^{L(t_n-\alpha)} = \frac{(\beta-\alpha)M}{2}\, e^{L(t_n-\alpha)}\, h.$$

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Result

$\|e_n\| = O(h)$ again: first-order convergence.

Remark: what the exponential factor means

Remark 2.5 (Grönwall bound versus direct unfolding: constants and the factor e^{Lt_n})

- Direct unfolding: constant $\frac{M}{2L}\left(e^{L(t_n-\alpha)}-1\right)$. Grönwall: constant $\frac{(\beta-\alpha)M}{2}e^{L(t_n-\alpha)}$. Both give order 1; the first is smaller.

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- $(1+hL)^n \le e^{Lt_n}$ for *every* h: the factor does not blow up when h is refined, it grows with the time t_n.

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- $(1+hL)^n \le e^{Lt_n}$ for *every* h: the factor does not blow up when h is refined, it grows with the time t_n.
- So for fixed t_n the bound holds for any h, but it is useless when $L(t_n-\alpha)$ is large. Stiff problems (L large) need other tools: implicit schemes and stability regions.

References

- **Grönwall inequality.** R. Howard, *The Gronwall inequality*, `arXiv:2503.23639` (ref.).

Q & A